

Zr Corrosion

$$T = 625 \text{ K}$$

$$t = 500 \text{ days}$$

$$\delta = 600 \text{ } \mu\text{m}$$

$$f = 0.14$$

$$\text{PBR} = 1.56$$

$$\rho_{\text{Zr}} = 6.5 \text{ g/cm}^3$$

$$\rho_{\text{ZrO}_2} = 5.68 \text{ g/cm}^3$$

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a. Oxide Thickness

$$t^* = 6.62 \times 10^{-7} e^{\frac{11949}{T}}$$

$$= 6.62 \times 10^{-7} e^{\frac{11949}{625 \text{ K}}}$$

$$= 133 \text{ days}$$

$$\delta^* = 5.1 e^{-\frac{550}{T}} = 5.1 e^{-\frac{550}{625}} = 2.115 \text{ } \mu\text{m}$$

$$K_2 = 7.48 \times 10^6 e^{-\frac{12500}{T}}$$

$$= 7.48 \times 10^6 e^{-\frac{12500}{625 \text{ K}}}$$

$$= 0.0154 \text{ } \mu\text{m/day}$$

$$\delta = \delta^* + K_2(t - t^*)$$

$$= 2.115 \text{ } \mu\text{m} + 0.0154 \frac{\mu\text{m}}{\text{day}} (500 \text{ day} - 133 \text{ day})$$

$$\delta = 7.7668 \text{ } \mu\text{m}$$

Fletcher  
Smith  
4-23-21

b. Wt ppm of hydrogen in cladding

$$C_{\text{H}}^{\text{clad}} (\text{wt. ppm}) = \frac{2 \cdot f \cdot \delta \cdot \rho_{\text{oxide}} \cdot f_{\text{ZrO}_2} \cdot \frac{M_{\text{H}}}{M_{\text{O}}}}{(t - \frac{\delta}{\text{PBR}}) \cdot \rho_{\text{metal}}} \times 10^6$$

Assuming 10 cm x 10 cm

$$M_{\text{O}} = \frac{m_{\text{O}} - m_1}{5} = \delta \cdot 14.7 = 7.7668 \cdot 14.7 = 114.2 \text{ mg}$$

$$M_{\text{H}} = 114.2 \text{ mg} \cdot \frac{1 \text{ g}}{1000 \text{ mg}} \cdot \frac{1 \text{ mol O}}{16 \text{ g O}} \cdot \frac{2 \text{ mol H}}{1 \text{ mol O}} \cdot \frac{1 \text{ g H}}{1 \text{ mol H}} \cdot \frac{100 \text{ mg}}{1 \text{ g}} = 28.55 \text{ mg H}$$

$$C_H^{\text{clad}} (\text{wt. ppm}) = \frac{2 \cdot f \cdot \delta \cdot \rho_{\text{oxide}} \cdot f_{\text{ZrO}_2}^{\text{O}} \cdot \frac{M_H}{M_O}}{\left(t - \frac{\delta}{\text{PBR}}\right) \cdot \rho_{\text{metal}}} \times 10^6$$

$$= \frac{2 \cdot 0.14 \cdot 7.7668 \mu\text{m} \cdot 5.68 \frac{\text{g}}{\text{cm}^3} \cdot 0.26 \cdot \frac{28.55 \text{ mg}}{114.2 \text{ mg}} \times 10^6}{\left(\overset{600}{500} - \frac{7.7668}{1.56}\right) \cdot 6.5 \text{ g/cm}^3}$$

$$C_H^{\text{clad}} = 249.5 \text{ wt. ppm}$$

- close...

2. The rate-limiting step in the aqueous corrosion is diffusion. Once an oxide layer forms on the Zirconium, oxygen ions need to diffuse through the oxide layer to get to bare Zr.  $\frac{d}{dt}$

3. Hydrides in the cladding tend to form in areas of high tensile stress (Soret effect) and low temperatures. ~~this is temperature~~  
Since hydrogen produced from oxidation travels so rapidly, hydrides can form anywhere throughout the fuel. They tend to align circumferentially during operation, then the hydrides realign radially when dried out. Hydrides are brittle and can cause blistering in the fuel, leading to ruptures.

4. RIA is Reactivity Insertion Accident. This typically happens in a PWR when control rods are suddenly extracted and in a BWR when control blades are suddenly dropped. During a RIA, a sudden increase in power causes an increase in heat generation and temperature. The fuel can expand and close the gap, inducing PCI. A crack may form and release fuel.

4. (continued) into the primary coolant. Fission gases may also be released from the fuel matrix and cause overpressurization inside the fuel rod. This can lead to ballooning and rupture of the fuel rod. Once fuel rods start rupturing the coolant channels may become blocked. This causes higher temperatures in the fuel. Similarly, the core may experience Departure from Nucleate Boiling, DNB, also leading to high temperatures. Ultimately the fuel may melt.

5. LOCA is a Loss Of Cooling Accident. In a LOCA, the core may experience a dry-out in a BWR or the introduction of some voids in a PWR. These will lead to not enough heat being removed and may induce some additional Fission Gas Release, impacting the integrity of the cladding with rupture and ballooning. DNB is possible as well as melt down. Many of the same phenomena as a RIA. The main difference is that a RIA happens very quickly and a LOCA occurs over much longer times than a RIA. The TH folks go nuts over LOCAs. LOCAs can also experience more steam generation and have an exothermic steam corrosion reaction with the cladding.

6. Burnup Impacts the fuel structure and thermal conductivity. In an accident scenario, fuels with a higher burnup are more likely to fracture due to a build up of pores and fission gas bubbles. They are already swelled as compared to fresh fuel, so PCI is already occurring. High burnup fuels have lower thermal conductivity and will release more fission gases.

- cladding embrittlement

7. Accident Tolerant Fuels seek to

1. Retain Fission Products in the fuel

2. Improve Fuel properties

3. Improve Cladding properties

4. Reduce Steam oxidation of cladding

These will help fuel perform/operate better at normal operating conditions and allow for more time to respond in an accident scenario. They should also be more resilient and not undergo accidents as easily.

8. One near-term ATF concept are ceramic coatings on the outside of the cladding. This should protect the cladding from steam oxidation and retain the ductility and structural integrity of the cladding better. A difficulty of this is the difference in coefficient of thermal expansion (CTE) is different. Other coatings that include Cr or Al tend to be very good at reducing corrosion.

9. When Zirconium is exposed to high-temperature steam, the steam will cause oxidation corrosion of the Zirconium. The temperature increases, and after the Zr increases in temperature past  $800^{\circ}\text{C}$ , it undergoes a phase transition from  $\alpha$  to  $\alpha + \beta$ , where the  $\alpha$  is stabilized by the oxide. It is very brittle and will rupture more easily.

10. 1. Power Generation ✓

5/6 2. Fuel centerline temperature - kind of

3. Departure from Nucleate Boiling ✓

4. Intergranular Stress Corrosion Cracking ✓

11. CRUD impacts the thermal conductivity of the cladding and inhibits heat from being removed from the fuel. It can also be deposited in other areas of the plant and cause undue radiation exposure to personnel in the plant.

12. pH control significantly lowers the rate at which Fe and Ni dissolve into the coolant and inhibits CRUD formation, and the addition of Zn and other noble metals causes CRUD to stay in the reactor by encouraging CRUD to form on the fuel rods. Kinda backwards, but it works. Tough balancing act.

13. 1. MOX fuels tend to have a large void at the center because voids diffuse up the thermal gradient to the center of the fuel and accumulate.

2. Equiaxed grains become columnar. Voids have one side where fuel is vaporizing and other side where fuel is condensing on cooler side, and as it travels towards center it leaves columnar grains behind.

- this is two parts of one phenomenon

